

Supplementary materials

MSYNC - Conversation Task

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S1 Introduction

This R Markdown script analyses Conversation Task data from the MSYNC project. The data was preprocessed before being read into this script. Here, we focus on the behavioural and the eye-tracking data. We use the following abbreviations:

- IPS : interpersonal synchrony between two interaction partners
- IPA : interpersonal adaptation, the amount one person adapts their behaviour to their interaction partner's behaviour
- QSE : quality of the social experience, collected through several self-reported measures

S1.1 Package versions

We used R version 4.5.2 (2025-10-31) and the following packages in the creation of this document: knitr version 1.51; brms version 2.23.0; tidyverse version 2.0.0; ggpubr version 1.0.0; bayesplot version 1.15.0; rstatix version 1.1.0; corrplot version 0.95; BayesFactor version 0.9.12.4.8; FactoMineR version 2.16; easystats version 0.7.6; psych version 2.6.5;

S1.2 General info

We performed prior predictive checks as proposed in Schad, Betancourt and Vasishth (2020) using the SBC package in a separate file. All categorical predictors are set to sum-coding and all continuous predictors and outcomes are scaled. Names of scaled variables are preceded by an s to indicate the scaling.

Table S1: Participant and dyad counts for different demo variables.

	cisgender	transgender	handedness	Freq	order	dyad.gender	n
-	0	1	links	6	CT-MG	female	5
female	44	0	rechts	74	CT-MG	male	2
male	34	1			CT-MG	mixed	13
					MG-CT	female	7
					MG-CT	male	5
					MG-CT	mixed	8

S1.3 Preparation and group comparisons

First, we load the data and combine it with demographic information including the diagnostic status of the subjects. Then, we print some basic information on the participants and dyads.

Table S2: Demographic information and questionnaire values

measurement	MG-CT	CT-MG	bf.log	overall
IRI_final	55.77 $\pm$ 8.17 (38 to 71)	56.85 $\pm$ 6.99 (40 to 72)	-1.286	56.31 $\pm$ 7.57 (38 to 72)
age	26.05 $\pm$ 3.59 (21 to 36)	25.38 $\pm$ 3.55 (19 to 37)	-1.170	25.71 $\pm$ 3.56 (19 to 37)
edu	4.92 $\pm$ 0.27 (4 to 5)	4.85 $\pm$ 0.36 (4 to 5)	-0.976	4.89 $\pm$ 0.32 (4 to 5)
ECR	41.73 $\pm$ 10.54 (19 to 61)	37.56 $\pm$ 11.18 (20 to 60)	-0.206	39.67 $\pm$ 10.99 (19 to 61)
ADC_final	31.62 $\pm$ 13.15 (6 to 62)	26.57 $\pm$ 12.40 (7 to 75)	0.421	29.10 $\pm$ 12.95 (6 to 75)
RAADS_final	66.20 $\pm$ 29.66 (6 to 111)	54.08 $\pm$ 29.12 (10 to 147)	0.940	60.14 $\pm$ 29.84 (6 to 147)
CFT_iq	110.10 $\pm$ 10.16 (93 to 126)	116.38 $\pm$ 12.52 (85 to 148)	1.194*	113.24 $\pm$ 11.76 (85 to 148)

The interaction partner in one dyad differed 3.08 ± 3.23 (0 to 11) years from each other.

S1.4 Assessing the relationship between motion quantity and motion IPSmov

Table S3: Credible correlations on dyad level for CT

Group	Parameter2	rho	CI_low	CI_high	pd	credible
CT-MG	MEA.mean	0.7724	0.3553	0.8946	1.0000	*
MG-CT	MEA.mean	0.6992	0.4204	0.8778	1.0000	*
MG-CT	MEA.zero	0.4744	0.1525	0.7685	0.9905	*

Table S4: Credible correlations on individual level for CT

Group	Parameter2	rho	CI_low	CI_high	pd	credible
CT-MG	MEA.mean	0.7399	0.6052	0.8593	1.0000	*
CT-MG	MEA.peak	0.3074	0.0399	0.5596	0.9802	*
MG-CT	MEA.mean	0.6361	0.4482	0.7998	1.0000	*
MG-CT	MEA.peak	0.3274	0.0435	0.5665	0.9882	*

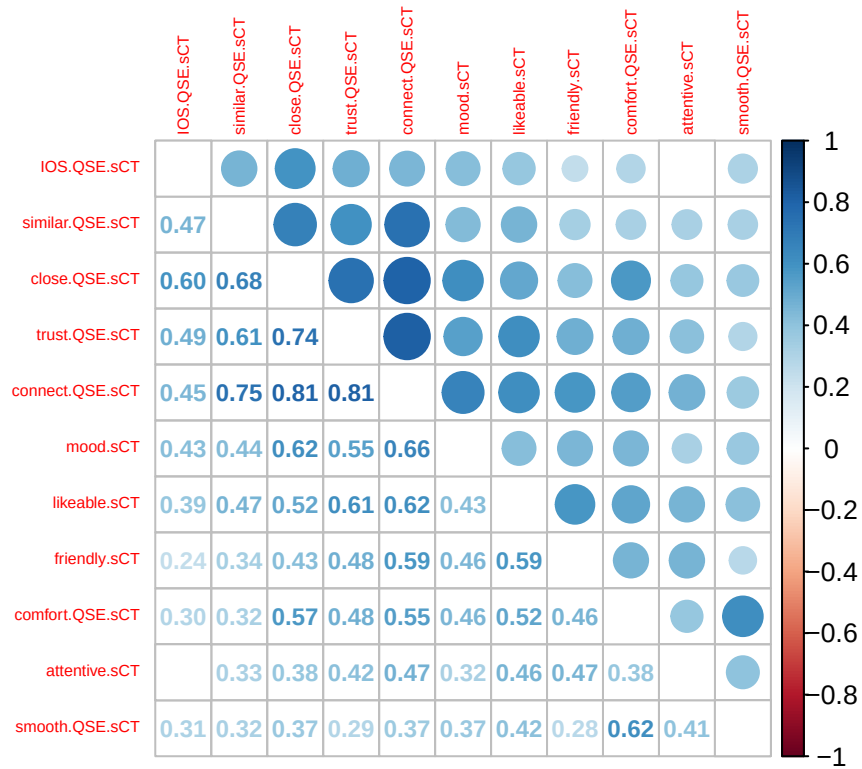
Since movement quantity and peak as well as average IPSmov/IPAmov are highly correlated in the conversation task, we will add this into our model as a predictor of no interest.

S1.5 Assessing the uniformity of QSE

We want to aggregate several scores from our questionnaires to measure the quality of the QSE. Thus, we check now if these scores are likely to measure one uniform concept.

```
## [1] "CT"
```

```
##
## A measure of unidimensionality
## Call: unidim(keys = select(df.indi, ends_with(".sCT")))
##
## Unidimensionality index =
##      u      tau      rho_c      alpha      av.r      median.r      CFI      ECV
##    0.91    0.92    0.99    0.91    0.47    0.46    0.88    0.79
##    F1/F2      MAP
##    4.99    0.04
##
## unidim adjusted index reverses negatively scored items.
## alpha      Based upon reverse scoring some items.
## average and median      correlations are based upon reversed scored items
```



S1.6 Setting weakly informed priors

We will use the same priors for all of our models except the Bernoulli model which got its own priors.

Table S5: Priors for all Gaussian linear (mixed) models

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
normal(0, 0.1)	Intercept						NA	NA		user
normal(1, 0.5)	sigma						NA	NA		user
normal(0.5, 0.5)	sd						NA	NA		user
normal(0, 0.3)	b						NA	NA		user

Table S6: Priors for the Bernoulli linear model

prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
normal(0, 0.5)	Intercept						NA	NA		user
normal(0, 1)	b						NA	NA		user

S2 Testing effect of order on IPSmov

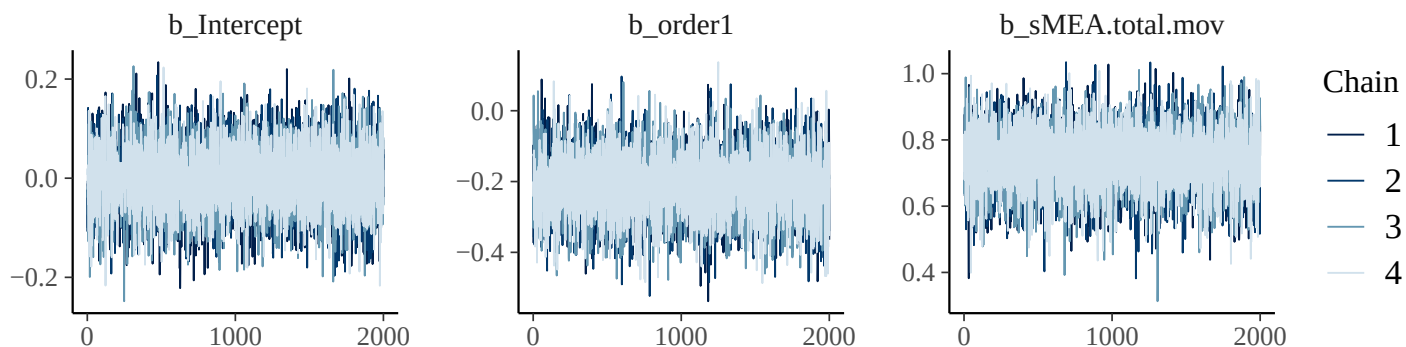
S2.1 Model setup

```
##      [,1]
## CT-MG    1
## MG-CT   -1
## sMEA.mean ~ order + sMEA.total.mov
```

S2.2 Posterior predictive checks

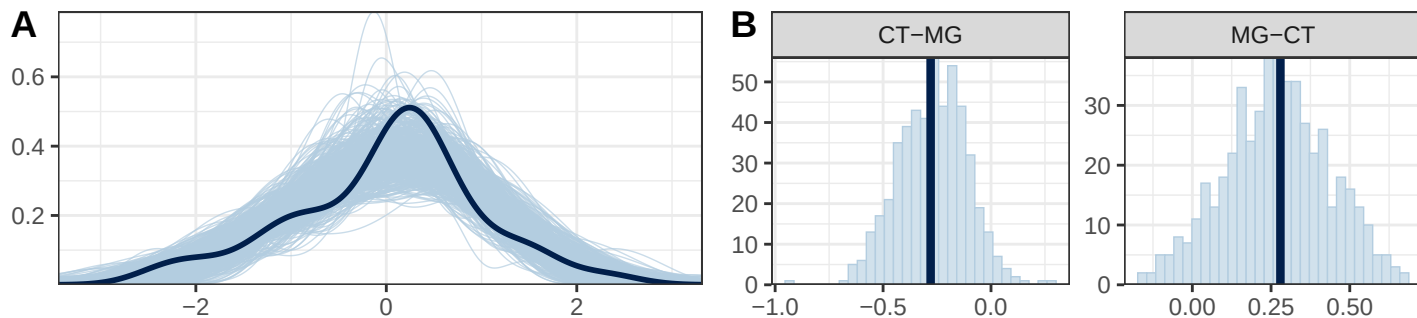
As the next step, we fit the model, check whether there are divergence or rhat issues, and then check whether the chains have converged.

```
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.
##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.
##
## Energy:
## E-BFMI indicated no pathological behavior.
## [1] 0
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



This model has no pathological behaviour with E-BFMI, no divergent sample and no rhats that are higher or equal to 1.01. Therefore, we go ahead and perform our posterior predictive checks.

Posterior predictive checks

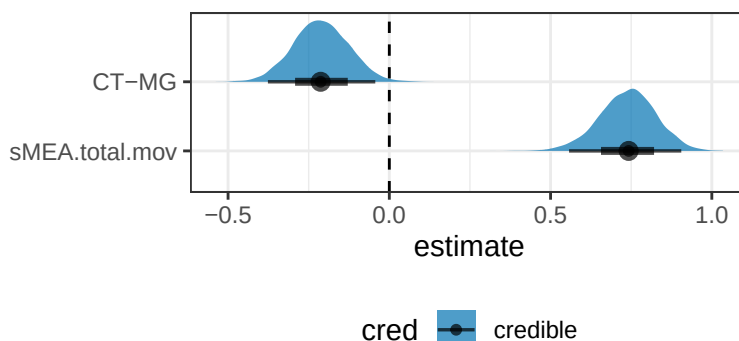


The overall simulated data (in lightblue) fits the real data (in dark blue) reasonably well, even though it does not reproduce the shape completely. The mean simulated data based on the model fits well with the real data.

S2.3 Inferences

Now that we are convinced that we can trust our model, we have a look at its estimates.

```
## Family: gaussian
## Links: mu = identity
## Formula: sMEA.mean ~ order + sMEA.total.mov
## Data: df.dyad.ct.mea (Number of observations: 40)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##       total post-warmup draws = 8000
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      -0.00     0.07   -0.13    0.13 1.00     9190     5520
## order1         -0.21     0.09   -0.38   -0.04 1.00     7621     5616
## sMEA.total.mov   0.74     0.09    0.56    0.91 1.00     8039     5459
##
## Further Distributional Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma         0.57     0.07    0.45    0.72 1.00     7816     5604
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

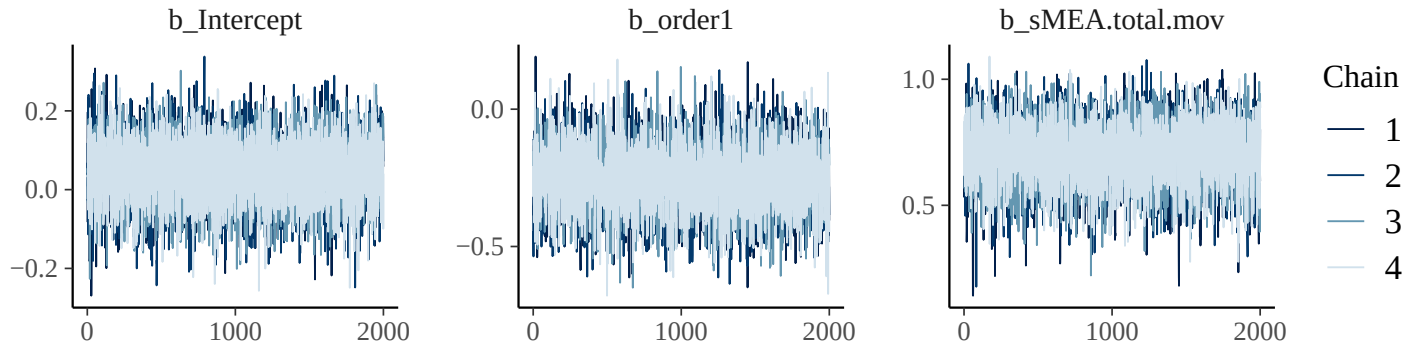


Assessing the effect of order on mean IPS results in the following $estimate = 0.42 [0.14, 0.71]$ (*posterior probability* = 99.15%); thus, this suggests a small effect of order on IPS of movement in the Conversation Task.

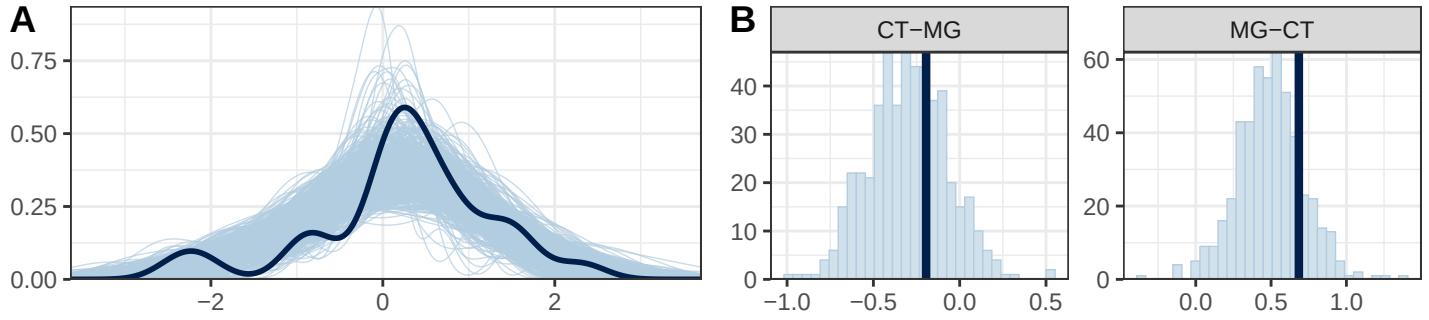
S2.4 Rerun focusing on dyads passing the pseudo-IPSmov test

Our comparison between pseudo-IPSmov and IPSmov did not reveal higher IPSmov across the board, but only for two-thirds of the sample. Therefore, we rerun the model on those dyads to ensure that the conclusions hold.

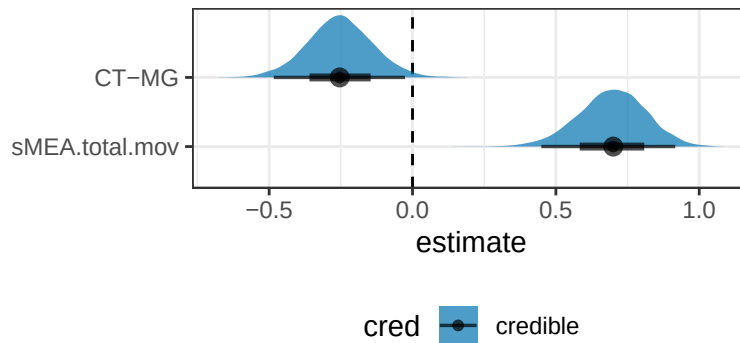
```
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.
##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.
##
## Energy:
## E-BFMI indicated no pathological behavior.
## [1] 0
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



Posterior predictive checks



```
## Family: gaussian
## Links: mu = identity
## Formula: sMEA.mean ~ order + sMEA.total.mov
## Data: df.sel (Number of observations: 26)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##         total post-warmup draws = 8000
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          0.04      0.08  -0.12   0.20 1.00     8501     6128
## order1             -0.25      0.11  -0.48  -0.03 1.00     6648     5207
## sMEA.total.mov       0.70      0.12   0.45   0.92 1.00     7139     4861
##
## Further Distributional Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma          0.61      0.10   0.45   0.85 1.00     6829     5594
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



Assessing the effect of order on mean IPS for these dyads only results in the following *estimate* = 0.51 [0.13, 0.88] (*posterior probability* = 98.52%); thus, this suggests a medium effect of order on IPS of movement in the Conversation Task when only considering dyads for which pseudo-IPS exceeded observed IPS.

S3 Exploring increasing social experience across the experiment

```
##      [,1]
## CT-MG    1
## MG-CT   -1

## sQSE.second ~ sQSE.first * order + (1 | dyad)

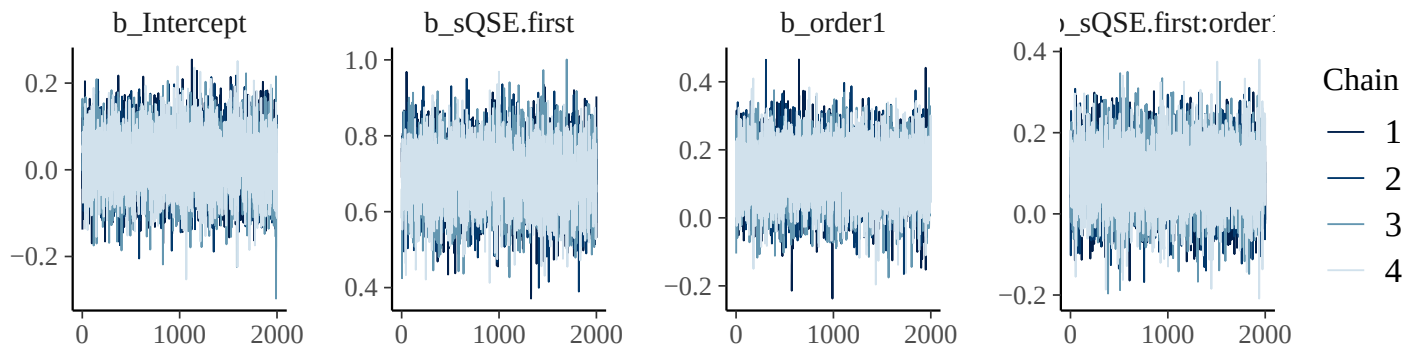
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.

##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.

##
## Energy:
## E-BFMI indicated no pathological behavior.

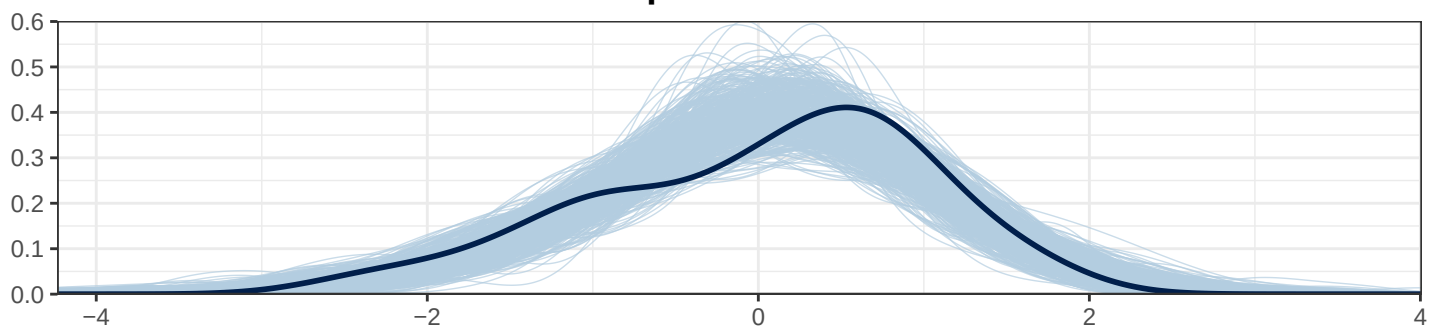
## [1] 0

## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



This model has no pathological behaviour with E-BFMI, no divergent sample and no rhats that are higher or equal to 1.01. Therefore, we go ahead and perform our posterior predictive checks.

Posterior predictive checks



The overall simulated data fits reasonably well. Now that we are convinced that we can trust our model, we have a look at its estimates.

```
## Family: gaussian
## Links: mu = identity
## Formula: sQSE.second ~ sQSE.first * order + (1 | dyad)
## Data: df.qse (Number of observations: 80)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~dyad (Number of levels: 40)
```

```
##          Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.32     0.13    0.05    0.56 1.00    1294    2132
##
## Regression Coefficients:
##          Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept        0.02     0.07   -0.11    0.15 1.00     7689     5887
## sQSE.first        0.69     0.08    0.53    0.85 1.00     4754     4987
## order1            0.13     0.08   -0.03    0.29 1.00     5444     4959
## sQSE.first:order1 0.09     0.08   -0.06    0.24 1.00     7527     6076
##
## Further Distributional Parameters:
##          Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma        0.58     0.07    0.47    0.72 1.00     2084     3687
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

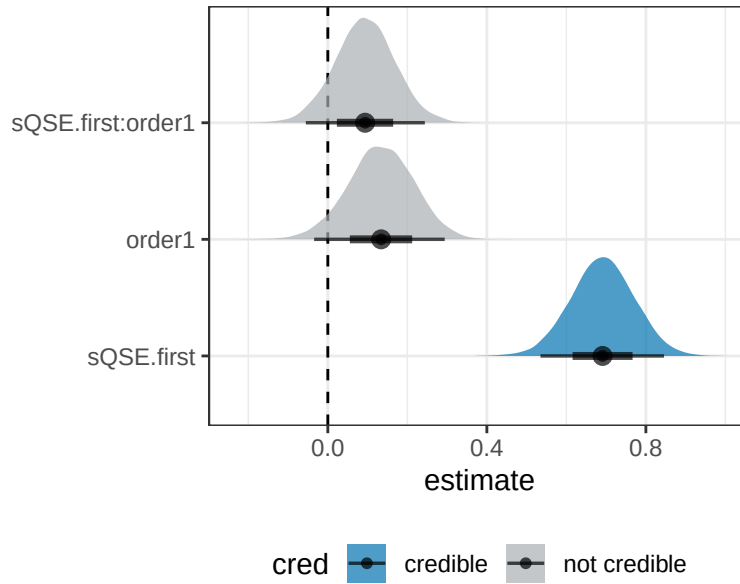
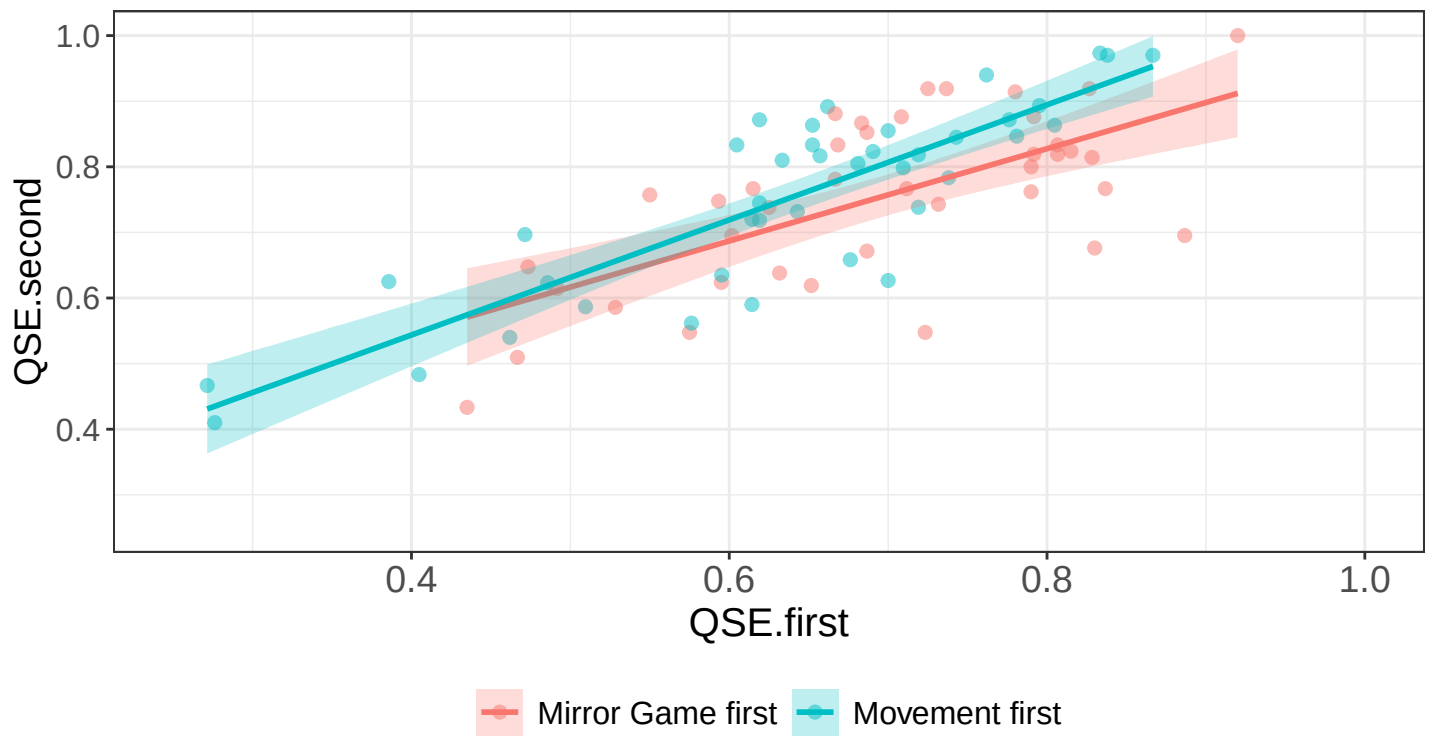


Table S7: Development of social experience across task

Effect	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
Order effect	$(\text{order1}^2) > 0$	0.267	0.167	-0.069	0.588	16.241	0.942	
Overall slope	$(\text{sQSE.first}) > 0$	0.691	0.080	0.535	0.846	Inf	1.000	*
CT-MG slope	$(\text{sQSE.first} + \text{sQSE.first:order1}) > 0$	0.785	0.105	0.575	0.989	Inf	1.000	*
MG-CT slope	$(\text{sQSE.first} - \text{sQSE.first:order1}) > 0$	0.597	0.115	0.370	0.824	Inf	1.000	*
Slope difference	$(2 * \text{sQSE.first:order1}) > 0$	0.187	0.151	-0.111	0.488	8.379	0.893	



S4 Exploring mediation analysis

S4.1 Model setup

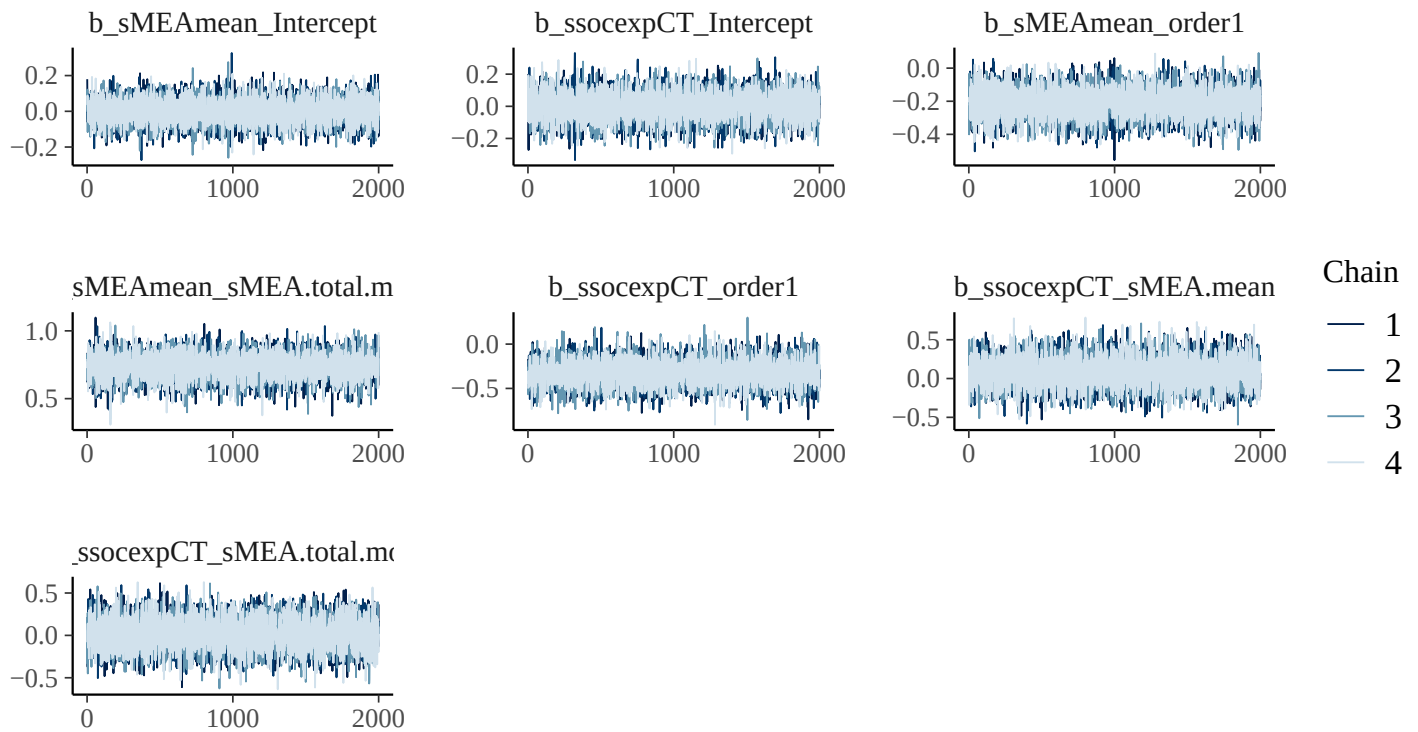
```
##          [,1]
## CT-MG      1
## MG-CT     -1

## sMEA.mean ~ order + sMEA.total.mov
## ssoc.exp.CT ~ order + sMEA.mean + sMEA.total.mov
```

S4.2 Posterior predictive checks

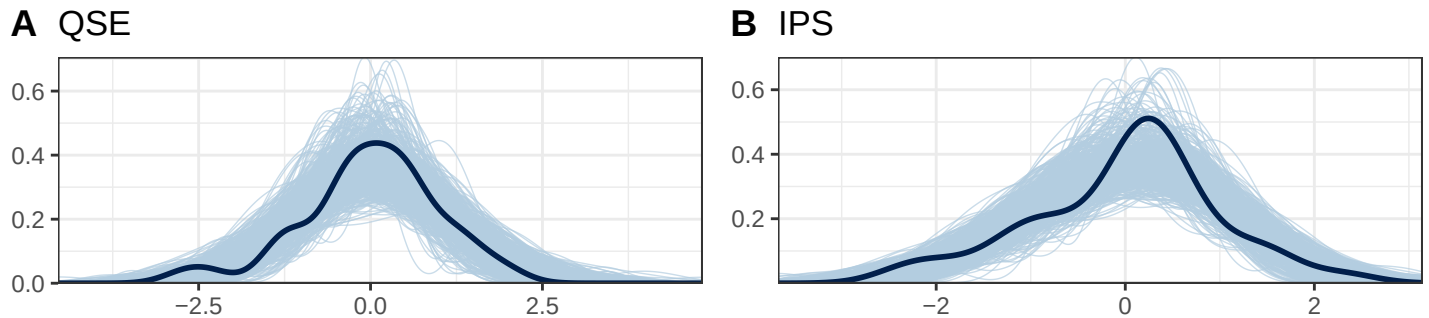
As the next step, we fit the model, check whether there are divergence or rhat issues, and then check whether the chains have converged.

```
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.
##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.
##
## Energy:
## E-BFMI indicated no pathological behavior.
## [1] 0
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



This model has no pathological behaviour with E-BFMI, no divergent sample and no rhats that are higher or equal to 1.01. Therefore, we go ahead and perform our posterior predictive checks.

Posterior predictive checks



The overall simulated data (in lightblue) fits the real data (in dark blue) reasonably well, even though it does not reproduce the shape completely. The mean simulated data based on the model fits well with the real data.

S4.3 Inferences

Now that we are convinced that we can trust our model, we have a look at its estimates.

```
## Family: MV(gaussian, gaussian)
## Links: mu = identity
##          mu = identity
## Formula: sMEA.mean ~ order + sMEA.total.mov
##          ssoc.exp.CT ~ order + sMEA.mean + sMEA.total.mov
## Data: df.ct.mea.qse (Number of observations: 40)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##              Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sMEAMean_Intercept      0.00    0.07  -0.13   0.13 1.00   11704
## ssocexpCT_Intercept     -0.00    0.08  -0.17   0.16 1.00   13574
## sMEAMean_order1        -0.21    0.08  -0.38  -0.04 1.00   12275
## sMEAMean_sMEA.total.mov  0.74    0.09   0.55   0.91 1.00   10924
## ssocexpCT_order1       -0.33    0.14  -0.60  -0.05 1.00   11237
```

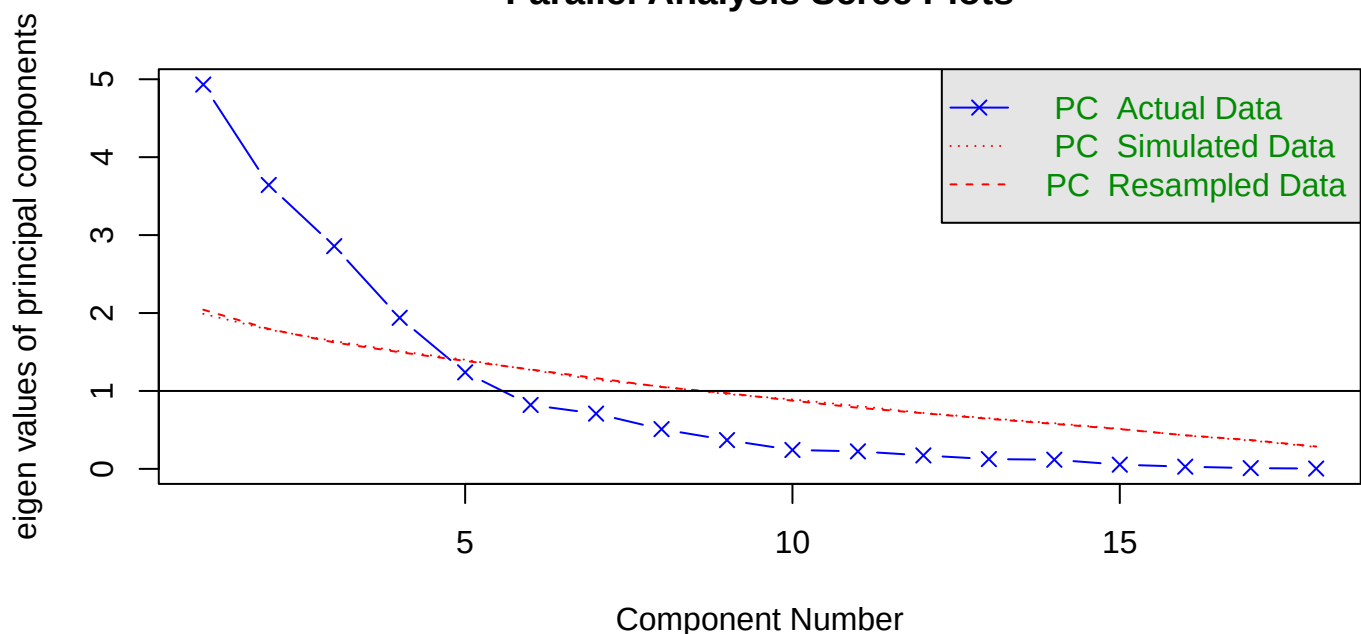
```
## ssocexpCT_sMEA.mean          0.09      0.18    -0.27      0.45 1.00      6997
## ssocexpCT_sMEA.total.mov     0.02      0.18    -0.33      0.37 1.00      7368
##                               Tail_ESS
## sMEAmean_Intercept          5876
## ssocexpCT_Intercept         5773
## sMEAmean_order1             6230
## sMEAmean_sMEA.total.mov     5915
## ssocexpCT_order1           5757
## ssocexpCT_sMEA.mean         6207
## ssocexpCT_sMEA.total.mov    6813
##
## Further Distributional Parameters:
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma_sMEAmean    0.56     0.07    0.45    0.72 1.00    10917    5822
## sigma_ssocexpCT   0.95     0.11    0.75    1.20 1.00    11462    6230
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Table S8: Results of the explorative mediation analysis

Parameter	Estimate	Posterior Probability	ROPE	ROPE Interpretation	HDI
IPS -> QSE.CT	0.090 +- 0.185 [-0.271, 0.447]	68.38	38.51	Undecided	[-0.27, 0.45]
Order -> IPS	0.423 +- 0.168 [0.089, 0.758]	99.35	0.34	Undecided	[0.09, 0.76]
Order -> IPS -> QSE.CT	0.038 +- 0.085 [-0.126, 0.221]	67.95	79.30	Undecided	[-0.13, 0.22]
Order -> QSE.CT	0.655 +- 0.277 [0.100, 1.197]	98.84	0.00	Rejected	[0.10, 1.20]
Total Effect	0.692 +- 0.269 [0.162, 1.218]	99.26	0.00	Rejected	[0.16, 1.22]

S5 Exploring individual experience and behaviour in CT: extracting principal components

Parallel Analysis Scree Plots



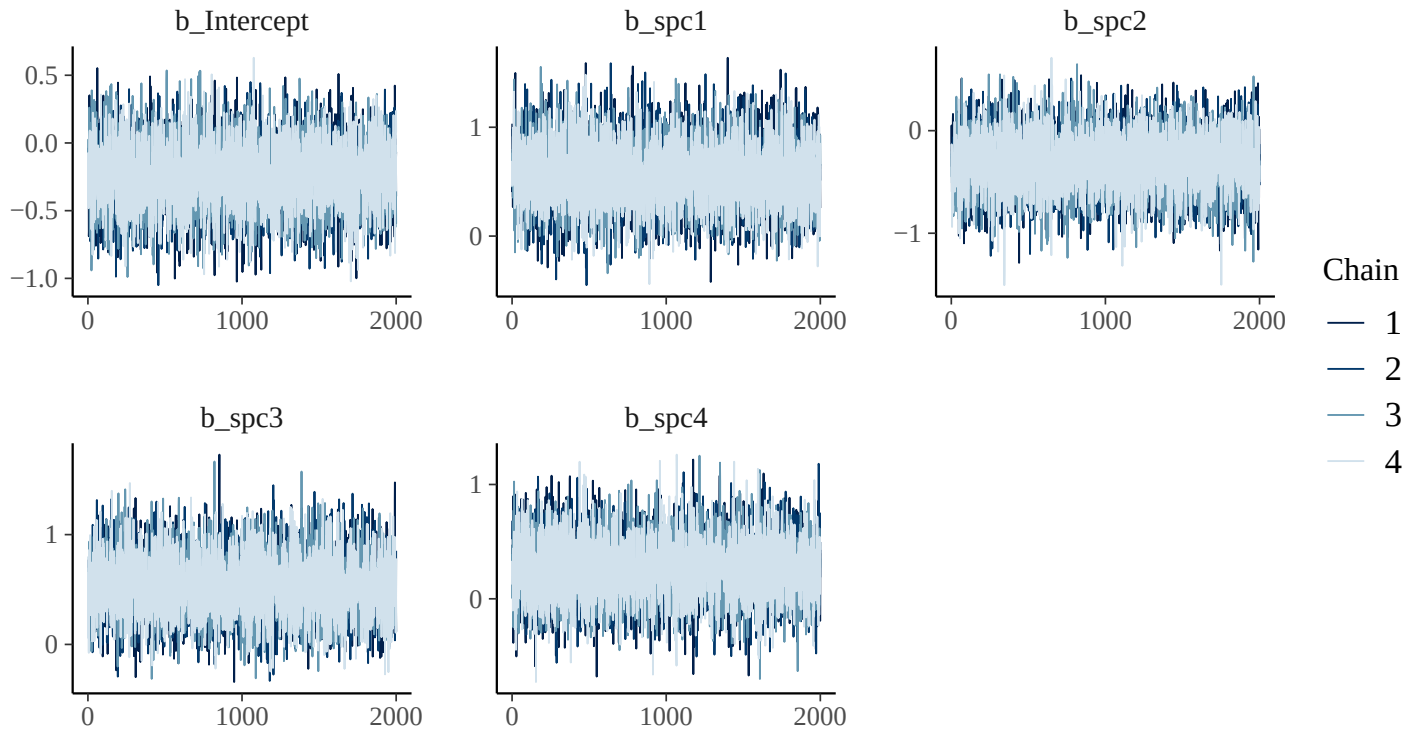
Parallel analysis suggests that the number of factors = NA and the number of components = 4

In total, there are 18 components, of which 4 are needed to explain at least 2/3 and 8 to explain more than 90% of the variance.

S5.1 Predicting order from PCs

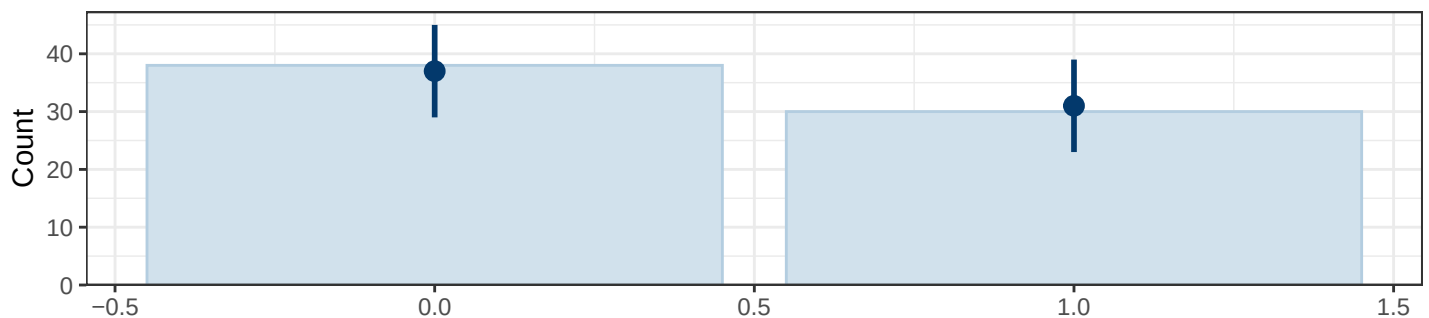
We try to predict order by the principal components. Since this is a model with a binary outcome we use a Bernoulli distribution with logit. We code the order CT-MG as 0 and the order MG-CT as 1. Thus, we need to use different priors here, because we are using a logit link function.

```
## order.bin ~ spc1 + spc2 + spc3 + spc4
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.
##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.
##
## Energy:
## E-BFMI indicated no pathological behavior.
## [1] 0
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



This model has no pathological behaviour with E-BFMI, no divergent sample and no rhats that are higher or equal to 1.01. Therefore, we go ahead and perform our posterior predictive checks.

Posterior predictive checks



The overall simulated data fits reasonably well. Now that we are convinced that we can trust our model, we have a look at its estimates.

```
## Family: bernoulli
## Links: mu = logit
## Formula: order.bin ~ spc1 + spc2 + spc3 + spc4
## Data: df.indi.ct.pc (Number of observations: 68)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept   -0.24      0.24   -0.72    0.22 1.00    9727    6377
## spc1         0.57      0.29    0.01    1.15 1.00    9000    6238
## spc2        -0.31      0.27   -0.86    0.22 1.00    9630    6298
## spc3         0.54      0.27    0.02    1.09 1.00   10203    6139
## spc4         0.26      0.26   -0.24    0.79 1.00    9882    6259
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

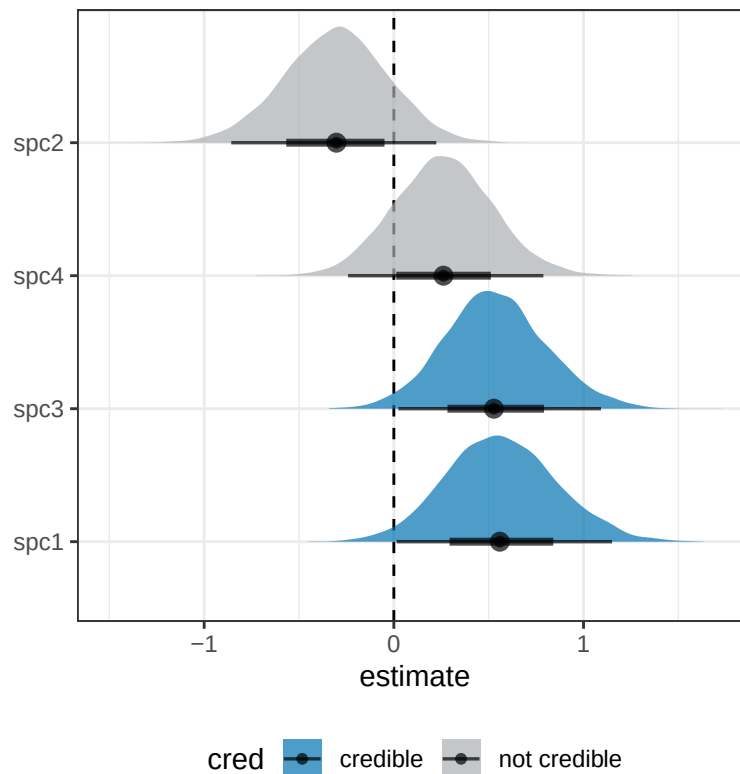


Table S9: Explorative assessment of Bernoulli model predictors
(ROPE Range = ± 0.181)

Parameter	Estimate	Posterior Probability	Effect size	ROPE	ROPE Interpretation	HDI
spc1	-0.567 ± 0.290 [-1.150, -0.015]	97.82*	medium	6.43	Undecided	[0.015, 1.150]
spc2	0.309 ± 0.275 [-0.224, 0.857]	87.24	small	30.13	Undecided	[-0.857, 0.224]
spc3	-0.536 ± 0.268 [-1.092, -0.024]	97.97*	medium	6.38	Undecided	[0.024, 1.092]
spc4	-0.263 ± 0.263 [-0.788, 0.241]	84.15	small	35.03	Undecided	[-0.241, 0.788]

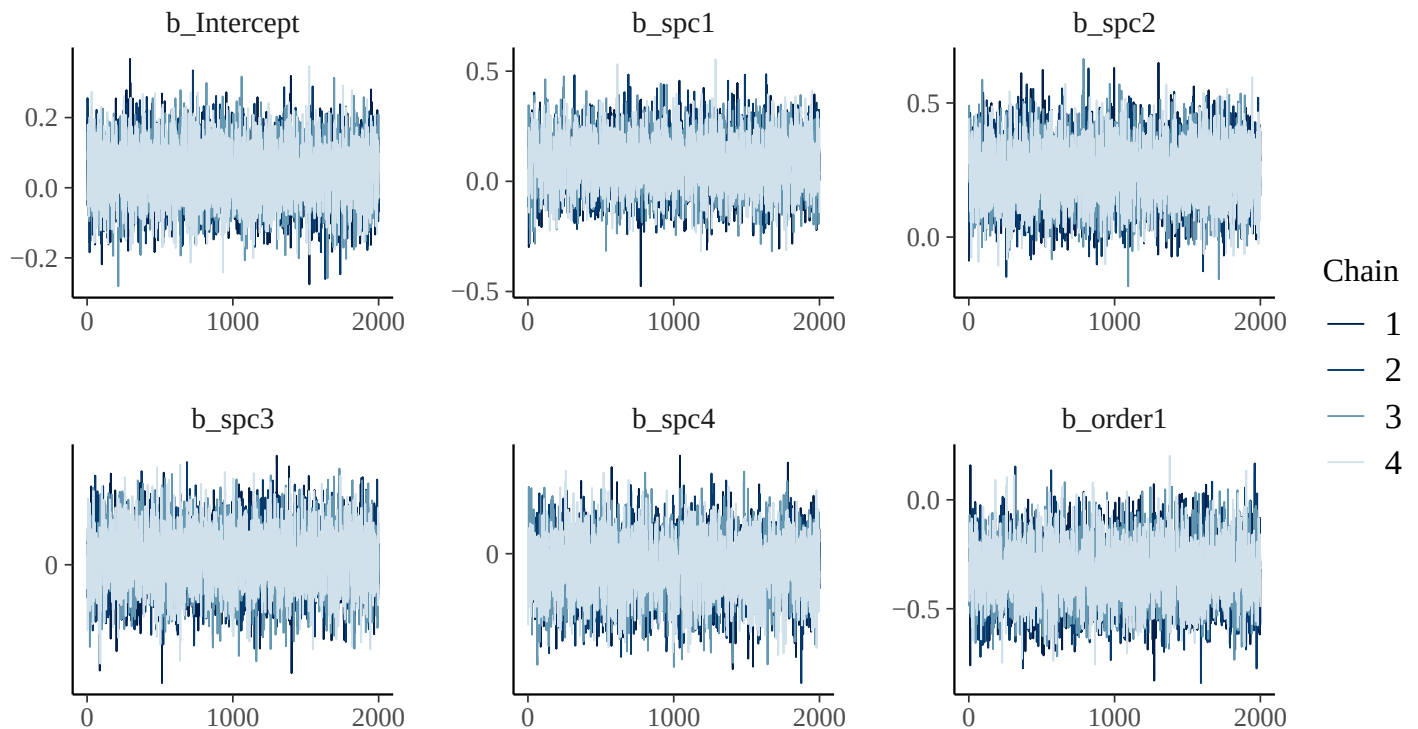
S5.2 Predicting social experience from PCs

```
##      [,1]
```

```
## CT-MG      1
## MG-CT     -1

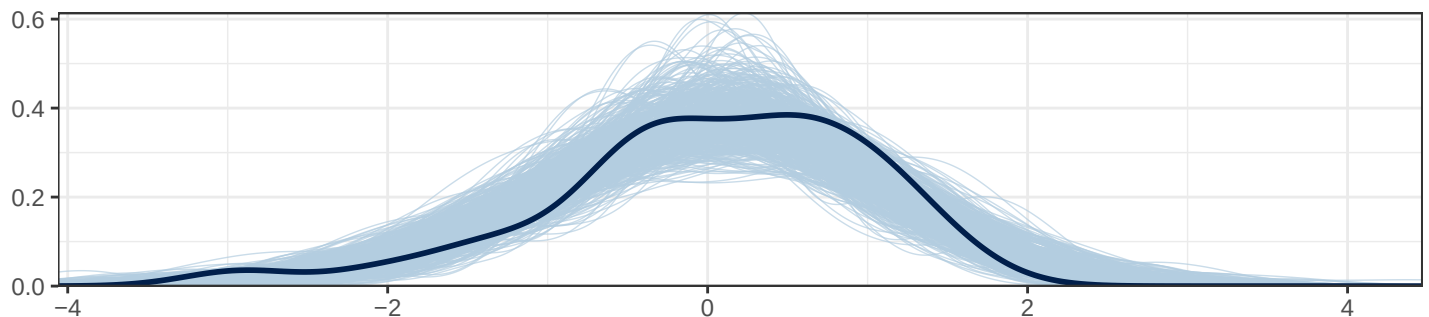
## ssoc.exp.CT ~ spc1 + spc2 + spc3 + spc4 + order + (1 | dyad)
##
## Divergences:
## 0 of 8000 iterations ended with a divergence.
##
## Tree depth:
## 0 of 8000 iterations saturated the maximum tree depth of 10.
##
## Energy:
## E-BFMI indicated no pathological behavior.
## [1] 0

## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



This model has no pathological behaviour with E-BFMI, no divergent sample and no rhats that are higher or equal to 1.01. Therefore, we go ahead and perform our posterior predictive checks.

Posterior predictive checks



The overall simulated data fits reasonably well. Now that we are convinced that we can trust our model, we have a look at its estimates.

```
## Family: gaussian
## Links: mu = identity
## Formula: ssoc.exp.CT ~ spc1 + spc2 + spc3 + spc4 + order + (1 | dyad)
## Data: df.indi.ct.pc (Number of observations: 68)
## Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
## total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~dyad (Number of levels: 34)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.67    0.13    0.41    0.95 1.00    2156    3097
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    0.04    0.08   -0.12    0.20 1.00    8144    6303
## spc1         0.08    0.12   -0.15    0.32 1.00    6141    5944
## spc2         0.24    0.11    0.02    0.46 1.00    6856    6514
## spc3         0.01    0.10   -0.18    0.22 1.00    8248    6296
## spc4        -0.05    0.10   -0.24    0.14 1.00    8781    6526
## order1      -0.33    0.14   -0.59   -0.06 1.00    5832    5701
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.64    0.09    0.50    0.84 1.00    2762    3951
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

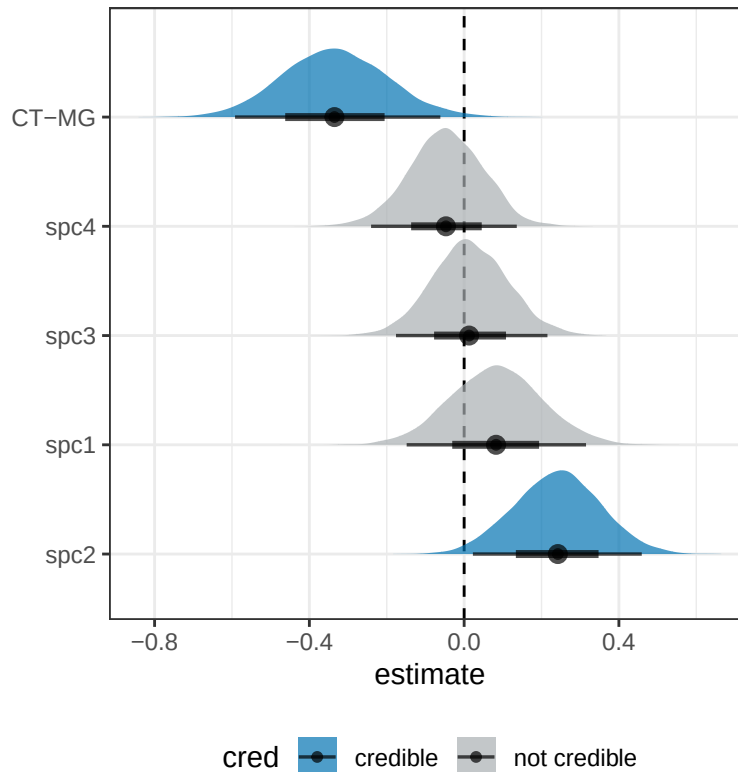


Table S10: Explorative assessment of Bernoulli model predictors
(ROPE Range = ± 0.100)

Parameter	Estimate	Posterior Probability	Effect size	ROPE	ROPE Interpretation	HDI
spc1	0.082 $\pm$ 0.118 [-0.149, 0.315]	75.41	very small	52.93	Undecided	[-0.149, 0.315]

Parameter	Estimate	Posterior Probability	Effect size	ROPE	ROPE Interpretation	HDI
spc2	0.241 $\pm$ 0.111 [0.023, 0.459]	98.56*	small	8.49	Undecided	[0.023, 0.459]
spc3	0.015 $\pm$ 0.099 [-0.176, 0.215]	55.31	very small	72.61	Undecided	[-0.176, 0.215]
spc4	-0.047 $\pm$ 0.096 [-0.241, 0.136]	68.59	very small	69.17	Undecided	[-0.241, 0.136]

S5.3 PC loadings

Table S11: All PCs together explaining >90%

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
MEA.total.mov	0.36	0.17	0.23	0.03	0.01	0.23	0.25	0.19
MEA.head.mov	0.34	0.17	0.22	0.05	0.00	0.19	0.03	-0.02
MEA.body.mov	0.34	0.16	0.24	0.00	0.03	0.22	0.30	0.22
MEA.mean	0.34	0.15	0.21	0.19	-0.04	-0.29	-0.01	-0.05
MEA.peak	0.17	0.19	0.20	0.29	-0.14	-0.63	-0.19	0.00
MEA.plag	0.06	-0.14	0.30	-0.35	-0.23	0.07	0.27	-0.72
OF.exp	0.15	0.13	0.23	-0.21	0.16	0.31	-0.82	-0.09
OF.mean_AU06_r	-0.14	0.44	0.01	-0.13	0.09	-0.14	0.00	-0.24
OF.mean_AU12_r	-0.18	0.42	-0.08	-0.20	0.06	0.09	0.09	0.15
OF.peak_AU06_r	-0.18	0.44	0.03	-0.11	0.09	-0.19	0.03	-0.20
OF.peak_AU12_r	-0.20	0.39	-0.04	-0.22	0.02	0.06	0.13	0.21
OF.plag_AU06_r	-0.01	-0.17	0.33	-0.46	0.03	-0.22	-0.05	0.20
OF.plag_AU12_r	-0.03	-0.23	0.21	-0.46	0.08	-0.33	0.07	0.34
dwel.face	0.28	-0.04	-0.27	-0.15	0.43	-0.11	0.00	0.00
FI	-0.03	-0.10	0.11	0.16	0.79	-0.08	0.17	-0.23
SI	0.32	0.00	-0.33	-0.18	0.14	-0.06	0.01	-0.05
SI.ratio	0.31	0.01	-0.36	-0.22	-0.10	-0.05	-0.06	-0.05
SI.FI	0.27	0.08	-0.36	-0.18	-0.18	-0.14	0.03	-0.10

Table S12: Most relevant PCs

	PC1	PC2	PC3	PC4
MEA.total.mov	0.36	0.17	0.23	
MEA.head.mov	0.34	0.17	0.22	
MEA.body.mov	0.34	0.16	0.24	
MEA.mean	0.34	0.15	0.21	0.19
MEA.peak	0.17	0.19	0.20	0.29
MEA.plag		-0.14	0.30	-0.35
OF.exp	0.15	0.13	0.23	-0.21
OF.mean_AU06_r	-0.14	0.44		-0.13
OF.mean_AU12_r	-0.18	0.42		-0.20
OF.peak_AU06_r	-0.18	0.44		-0.11
OF.peak_AU12_r	-0.20	0.39		-0.22
OF.plag_AU06_r		-0.17	0.33	-0.46
OF.plag_AU12_r		-0.23	0.21	-0.46
dwel.face	0.28		-0.27	-0.15
FI			0.11	0.16
SI	0.32		-0.33	-0.18
SI.ratio	0.31		-0.36	-0.22
SI.FI	0.27		-0.36	-0.18